Unit 4 Seminar: Linear Regression with Scikit-Learn

Introduction

The fuel-consumption demo and both global-data tasks were completed. Population alone showed little linear association with mean GDP per person across the analysed economies.

Data preparation

The supplied GDP records total output and end in 2020. Annual GDP was divided by same-year population before averaging. Codes were joined after resolving conflicting Aruba rows by coverage. A separate World Bank snapshot supplied 2001–2021 without mixing vintages; 49 aggregates and 16 economies lacking complete annual pairs were excluded (World Bank, 2026a).

Findings

The corrected fuel model was predicted from engine size, using a seeded split and the proper R^2 argument order (scikit-learn developers, no date). Test R^2 was 0.778 and MAE 22.58 g/km. Related vehicle variants may cross the split, limiting generalisation. The supplementary China sigmoid demonstrates curve fitting, without forecasting validation.

Correlation and regression

Across 201 complete-period economies, Pearson's r was −0.0667 and in-sample R^2 0.00445. The fitted slope was −11.66 US dollars per person per million residents. Residuals and the near-horizontal fit show limited usefulness; available-year and large-country sensitivity checks also produced weak linear associations. The separate holdout R^2 was −0.038. Following Géron (2025), I distinguish descriptive fit from predictive evaluation.

Implications and conclusion

Complete-period selection may favour better-reported economies. Current-dollar averages also reflect inflation and exchange rates. National GDP per person should not be used to label individual prosperity or justify discriminatory decisions. Professionally, source definitions, joins and uncertainty require checking before modelling; legally, World Bank attribution and licensing conditions must be retained (World Bank, 2026b). The exercise supports cautious dataset preparation and interpretation rather than a causal population claim.

Narrative 250 words excluding headings, evidence tables, figure captions and references. Execution was local; the notebook is compatible with Google Colab.

Fuel consumption evidence

Measure	Verified result
Records and seed	1,067 vehicles; 848 training / 219 test; seed 42
Engine size fit	CO2 = 124.176 + 39.526 × engine size in litres
Held-out errors	MAE 22.577 g/km; MSE 851.048 (g/km)²; RMSE 29.173 g/km
Held-out score	R^2 0.77819; training-mean baseline MAE 51.497 g/km
Full-table Pearson r	Engine size 0.8742; cylinders 0.8497; combined L/100 km 0.8921 with CO2

Table 1: Results from FuelConsumption.csv. Correlations use all rows; performance uses held-out rows. The seeded mask gives an approximate 80/20 split.

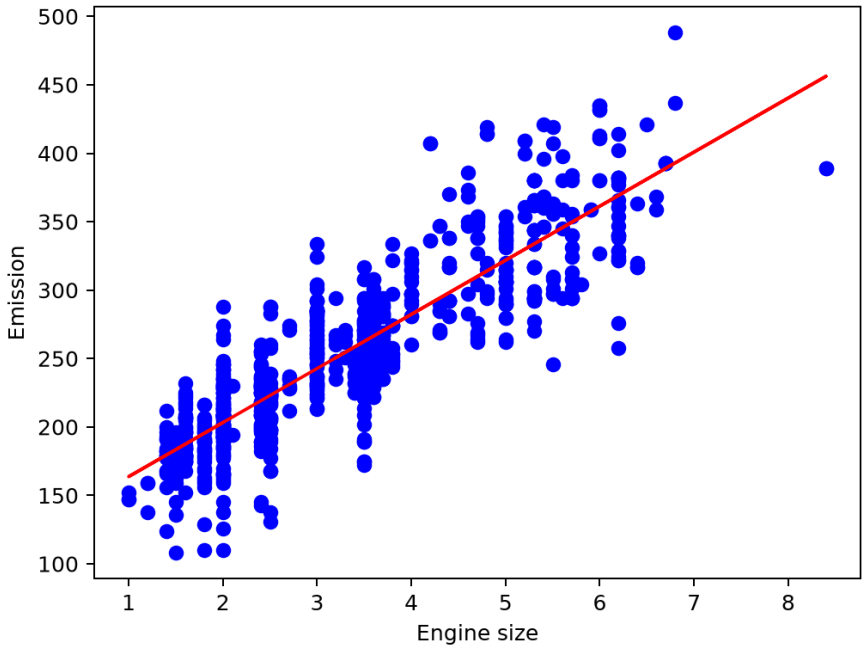


Figure 1: Training data and fitted engine-size regression. The coefficient describes the association in the supplied 2014 table.

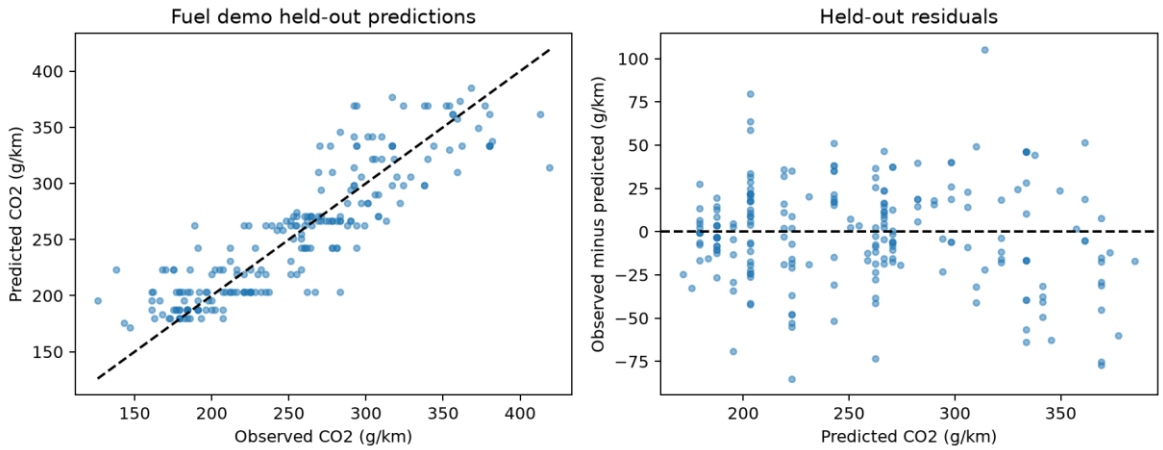


Figure 2: Held-out predictions and residuals. Imperfect predictions remain despite the positive test R^2.

Population and GDP evidence

Analysis	n	Pearson r	In-sample R^2
Supplied 2001–2020 complete	185	−0.04947	0.00245
Refreshed 2001–2021 complete	201	−0.06668	0.00445
Refreshed available years	214	−0.07044	0.00496
Refreshed without China and India	199	−0.05894	0.00347

Table 2: Arithmetic means of annual GDP/population ratios and matching population values. Main cohort requires 21 valid annual pairs. The supplied cohort requires 20. Differences between rows can reflect coverage, vintages and periods; they are not isolated effects of adding 2021.

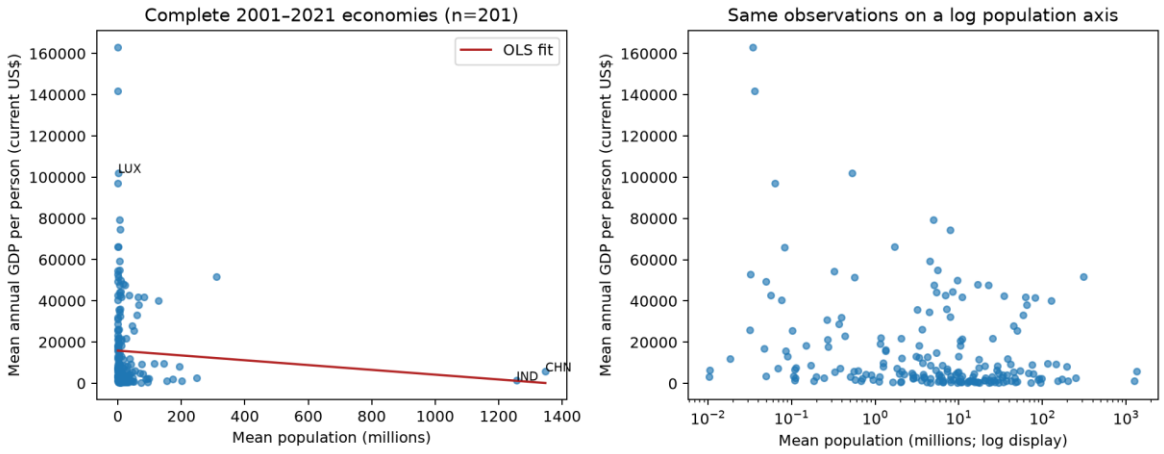


Figure 3: Main 2001–2021 analysis. The OLS line uses raw population; the right panel changes only the display scale. GDP is measured in current US dollars per person.

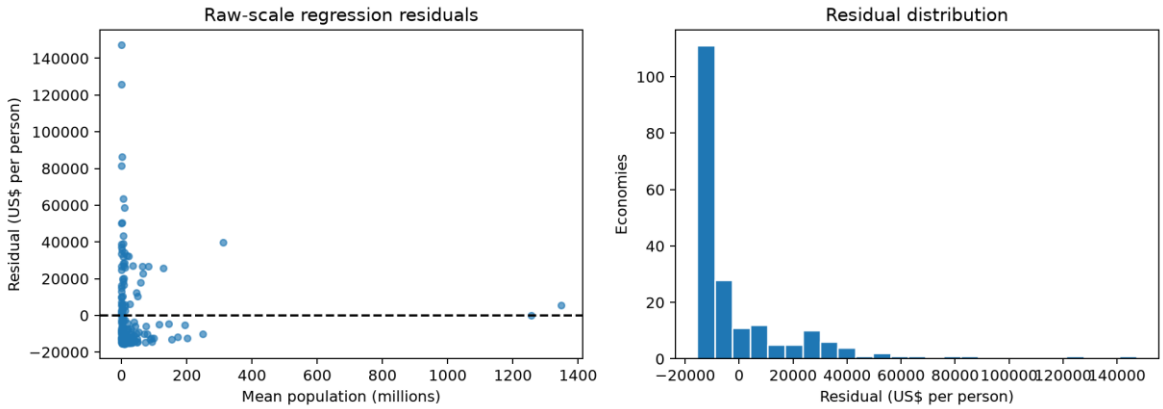


Figure 4: Residual spread and asymmetry caution against treating the line as a useful general model. Main fit: intercept US$15,830.88; slope −US$11.656 per million residents.

Predictive check: 160 training and 41 test economies, seed 42. Test R^2 −0.03767; MAE US$14,530.86 per person, compared with US$14,860.19 for a training-mean prediction. Negative R^2 and slightly lower MAE can coexist because they use different loss functions.

Supplementary demo and references

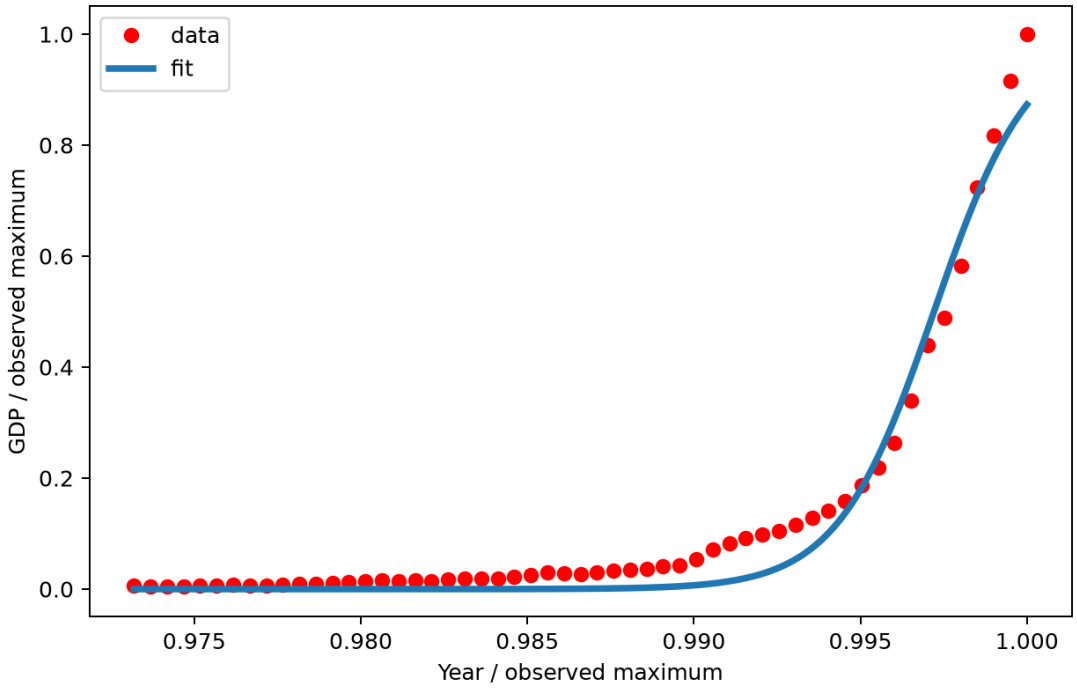


Figure 5: Supplied notebook's supplementary nonlinear section, using IBM's 55 annual China GDP teaching observations, 1960–2014. Both axes are normalised by observed maxima. This in-sample sigmoid fit is not Task B and does not validate forecasts.

References

Géron, A. (2025) *Hands-On Machine Learning with Scikit-Learn and PyTorch*. Santa Rosa: O'Reilly.

IBM (no date) *China GDP Teaching Dataset* [Dataset]. Available at: https://s3-api.us-geo.objectstorage.softlayer.net/cf-courses-data/CognitiveClass/ML0101ENv3/labs/china_gdp.csv (Accessed: 24 October 2026).

scikit-learn developers (no date) *r2_score*. Available at: https://scikit-learn.org/stable/modules/generated/sklearn.metrics.r2_score.html (Accessed: 24 October 2026).

World Bank (2026a) *World Development Indicators: GDP (current US$), population and economy metadata* [Dataset]. Indicator codes NY.GDP.MKTP.CD and SP.POP.TOTL. Data update 8 October 2026. Available at: https://data.worldbank.org/indicator/NY.GDP.MKTP.CD and https://data.worldbank.org/indicator/SP.POP.TOTL (Accessed: 24 October 2026).

World Bank (2026b) *GDP (current US$): licence CC BY 4.0*. Available at: https://data.worldbank.org/indicator/NY.GDP.MKTP.CD (Accessed: 24 October 2026).

Source artefacts: University of Essex Unit 4 Seminar brief, supplied demo notebook, FuelConsumption.csv, Global_GDP.csv and Global_Population.csv. Full snapshot URLs, package versions, source hashes, tables and execution evidence accompany the notebook.

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